

Do **not** write solutions on this page.

Section B

Answer **all** questions in the answer booklet provided. Please start each question on a new page.

10. [Maximum mark: 15]

A shop sells chocolates. The weight, in kilograms, of chocolates bought by a random customer can be modelled by a continuous random variable X with probability density function f defined by

$$f(x) = \begin{cases} \frac{6}{85}(4 + 3x - x^2), & 0.5 \leq x \leq 3 \\ 0, & \text{otherwise} \end{cases}.$$

- (a) Find the mode of X . [2]
- (b) Find $P(1 \leq X \leq 2)$. [2]
- (c) Find the median of X . [3]

The shop sells chocolates to customers at \$25 per kilogram.

However, if the weight of chocolate bought by a customer is at least 0.75 kilograms, the shop sells chocolate at a discounted rate of \$24 per kilogram.

- (d) Find the probability that a randomly selected customer spends at most \$48. [3]
- (e) Find the expected amount spent per customer. Give your answer correct to the nearest cent. [5]

